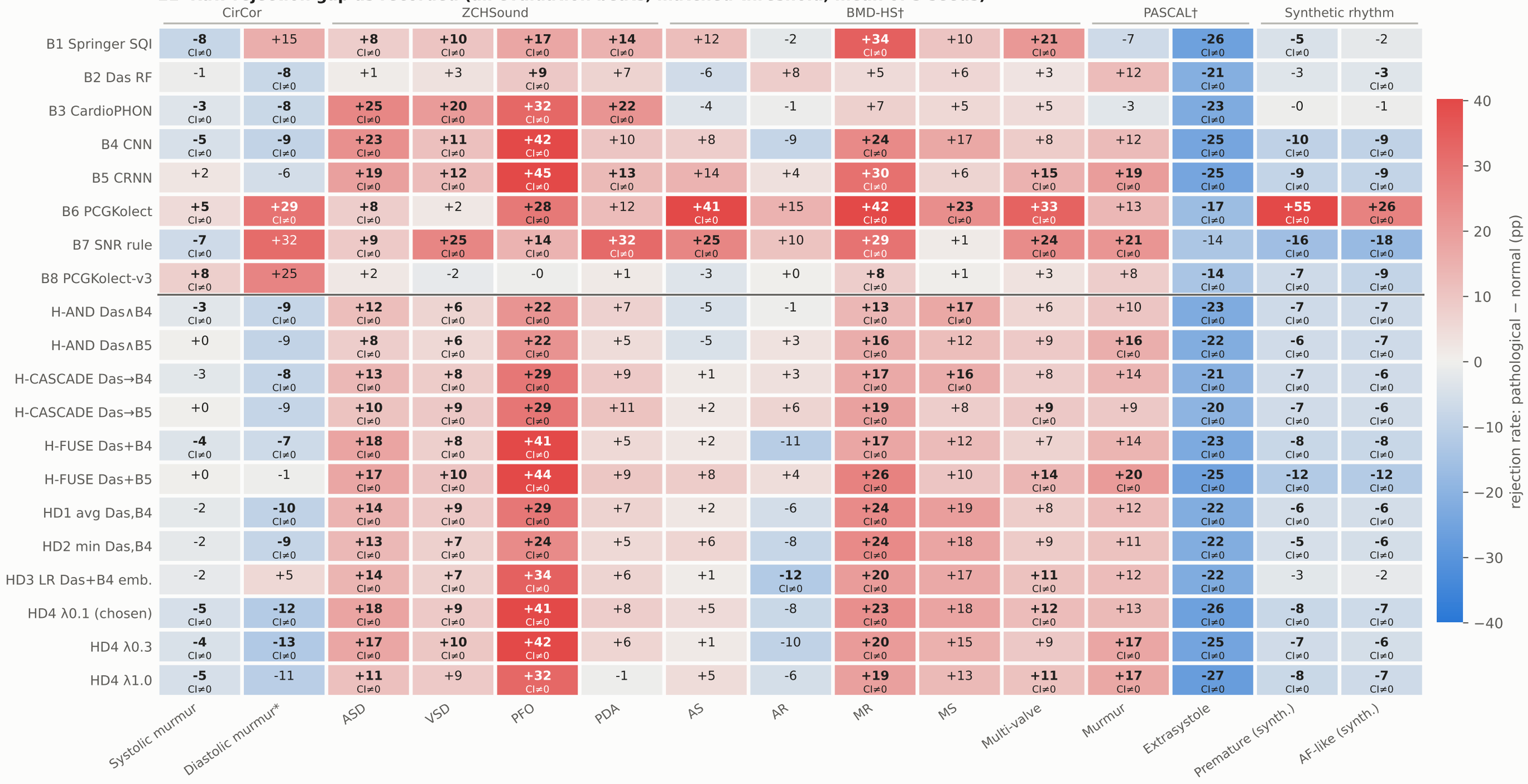
E1 Raw rejection gap as recorded (all evaluation beats, matched threshold, mean of 3 seeds)



	Systolic murmur	Diastolic murmur*	ASD	VSD	PFO	PDA	AS	AR	MR	MS	Multi-valve	Murmur	Extrasystole	Premature (synth.)	AF-like (synth.)
B1 Springer SQI	**-8** CI≠0	+15	**+8** CI≠0	**+10** CI≠0	**+17** CI≠0	**+14** CI≠0	+12	-2	**+34** CI≠0	+10	**+21** CI≠0	-7	**-26** CI≠0	**-5** CI≠0	-2
B2 Das RF	-1	**-8** CI≠0	+1	+3	**+9** CI≠0	+7	-6	+8	+5	+6	+3	+12	**-21** CI≠0	-3	**-3** CI≠0
B3 CardioPHON	**-3** CI≠0	**-8** CI≠0	**+25** CI≠0	**+20** CI≠0	**+32** CI≠0	**+22** CI≠0	-4	-1	+7	+5	+5	-3	**-23** CI≠0	-0	-1
B4 CNN	**-5** CI≠0	**-9** CI≠0	**+23** CI≠0	**+11** CI≠0	**+42** CI≠0	+10	+8	-9	**+24** CI≠0	+17	+8	+12	**-25** CI≠0	**-10** CI≠0	**-9** CI≠0
B5 CRNN	+2	-6	**+19** CI≠0	**+12** CI≠0	**+45** CI≠0	**+13** CI≠0	+14	+4	**+30** CI≠0	+6	**+15** CI≠0	**+19** CI≠0	**-25** CI≠0	**-9** CI≠0	**-9** CI≠0
B6 PCGKolect	**+5** CI≠0	**+29** CI≠0	**+8** CI≠0	+2	**+28** CI≠0	+12	**+41** CI≠0	+15	**+42** CI≠0	**+23** CI≠0	**+33** CI≠0	+13	**-17** CI≠0	**+55** CI≠0	**+26** CI≠0
B7 SNR rule	**-7** CI≠0	+32	**+9** CI≠0	**+25** CI≠0	**+14** CI≠0	**+32** CI≠0	**+25** CI≠0	+10	**+29** CI≠0	+1	**+24** CI≠0	**+21** CI≠0	-14	**-16** CI≠0	**-18** CI≠0
B8 PCGKolect-v3	**+8** CI≠0	+25	+2	-2	-0	+1	-3	+0	**+8** CI≠0	+1	+3	+8	**-14** CI≠0	**-7** CI≠0	**-9** CI≠0
H-AND Das∧B4	**-3** CI≠0	**-9** CI≠0	**+12** CI≠0	**+6** CI≠0	**+22** CI≠0	+7	-5	-1	**+13** CI≠0	**+17** CI≠0	+6	+10	**-23** CI≠0	**-7** CI≠0	**-7** CI≠0
H-AND Das∧B5	+0	-9	**+8** CI≠0	**+6** CI≠0	**+22** CI≠0	+5	-5	+3	**+16** CI≠0	+12	+9	**+16** CI≠0	**-22** CI≠0	**-6** CI≠0	**-7** CI≠0
H-CASCADE Das→B4	-3	**-8** CI≠0	**+13** CI≠0	**+8** CI≠0	**+29** CI≠0	+9	+1	+3	**+17** CI≠0	**+16** CI≠0	+8	+14	**-21** CI≠0	**-7** CI≠0	**-6** CI≠0
H-CASCADE Das→B5	+0	-9	**+10** CI≠0	**+9** CI≠0	**+29** CI≠0	+11	+2	+6	**+19** CI≠0	+8	**+9** CI≠0	+9	**-20** CI≠0	**-7** CI≠0	**-6** CI≠0
H-FUSE Das+B4	**-4** CI≠0	**-7** CI≠0	**+18** CI≠0	**+8** CI≠0	**+41** CI≠0	+5	+2	-11	**+17** CI≠0	+12	+7	+14	**-23** CI≠0	**-8** CI≠0	**-8** CI≠0
H-FUSE Das+B5	+0	-1	**+17** CI≠0	**+10** CI≠0	**+44** CI≠0	+9	+8	+4	**+26** CI≠0	+10	**+14** CI≠0	**+20** CI≠0	**-25** CI≠0	**-12** CI≠0	**-12** CI≠0
HD1 avg Das,B4	-2	**-10** CI≠0	**+14** CI≠0	**+9** CI≠0	**+29** CI≠0	+7	+2	-6	**+24** CI≠0	+19	+8	+12	**-22** CI≠0	**-6** CI≠0	**-6** CI≠0
HD2 min Das,B4	-2	**-9** CI≠0	**+13** CI≠0	**+7** CI≠0	**+24** CI≠0	+5	+6	-8	**+24** CI≠0	+18	+9	+11	**-22** CI≠0	**-5** CI≠0	**-6** CI≠0
HD3 LR Das+B4 emb.	-2	+5	**+14** CI≠0	**+7** CI≠0	**+34** CI≠0	+6	+1	**-12** CI≠0	**+20** CI≠0	+17	**+11** CI≠0	+12	**-22** CI≠0	-3	-2
HD4 λ0.1 (chosen)	**-5** CI≠0	**-12** CI≠0	**+18** CI≠0	**+9** CI≠0	**+41** CI≠0	+8	+5	-8	**+23** CI≠0	+18	**+12** CI≠0	+13	**-26** CI≠0	**-8** CI≠0	**-7** CI≠0
HD4 λ0.3	**-4** CI≠0	**-13** CI≠0	**+17** CI≠0	**+10** CI≠0	**+42** CI≠0	+6	+1	-10	**+20** CI≠0	+15	+9	**+17** CI≠0	**-25** CI≠0	**-7** CI≠0	**-6** CI≠0
HD4 λ1.0	**-5** CI≠0	-11	**+11** CI≠0	+9	**+32** CI≠0	-1	+5	-6	**+19** CI≠0	+13	**+11** CI≠0	**+17** CI≠0	**-27** CI≠0	**-8** CI≠0	**-7** CI≠0

Red = pathology rejected more often than normal beats. Bold/"CI≠0" = 95 % patient-bootstrap CI excludes 0 in ≥ 2 of 3 seeds. * 5 patients. † small calibration sets. E1 mixes recording quality and filter behaviour; E2 isolates the filter. Rows below the line: hybrids.